AI-Driven Smart Fashion Retail Recommendation System

Final Presentation · Group R26-IT-060

BSc (Hons) Information Technology · SLIIT · October 2026

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PP2 comments and remarks

Topic	AI Based Smart Retail Recommendation System			
Project ID	R26-IT-060			
[illegible]	IT			
Student ID	IT22146588	IT22244598	IT22284952	IT22243812
PanelName	IT-P05			
Detailed Comments	Very Good Progress for PP2 milestone.	Good Progress for PP2 milestone.	Very Good Progress for PP2 milestone.	Very Good Progress for PP2 milestone.

Office on the web Frame

Problem in brief: retail decisions are made in silos

Multi-store fashion retailers run inventory, promotion, fitting and in-store service on tools that don't share data — so one decision never informs the others.

1

Reactive inventory

Stock forecast and transferred manually, ignoring promotions, seasonality and weather

2

Stock-unaware marketing

Broad segments, no redemption likelihood — campaigns can push out-of-stock items

3

Fit uncertainty

Sizes differ across brands; try-on tools separate measurement from visualization

4

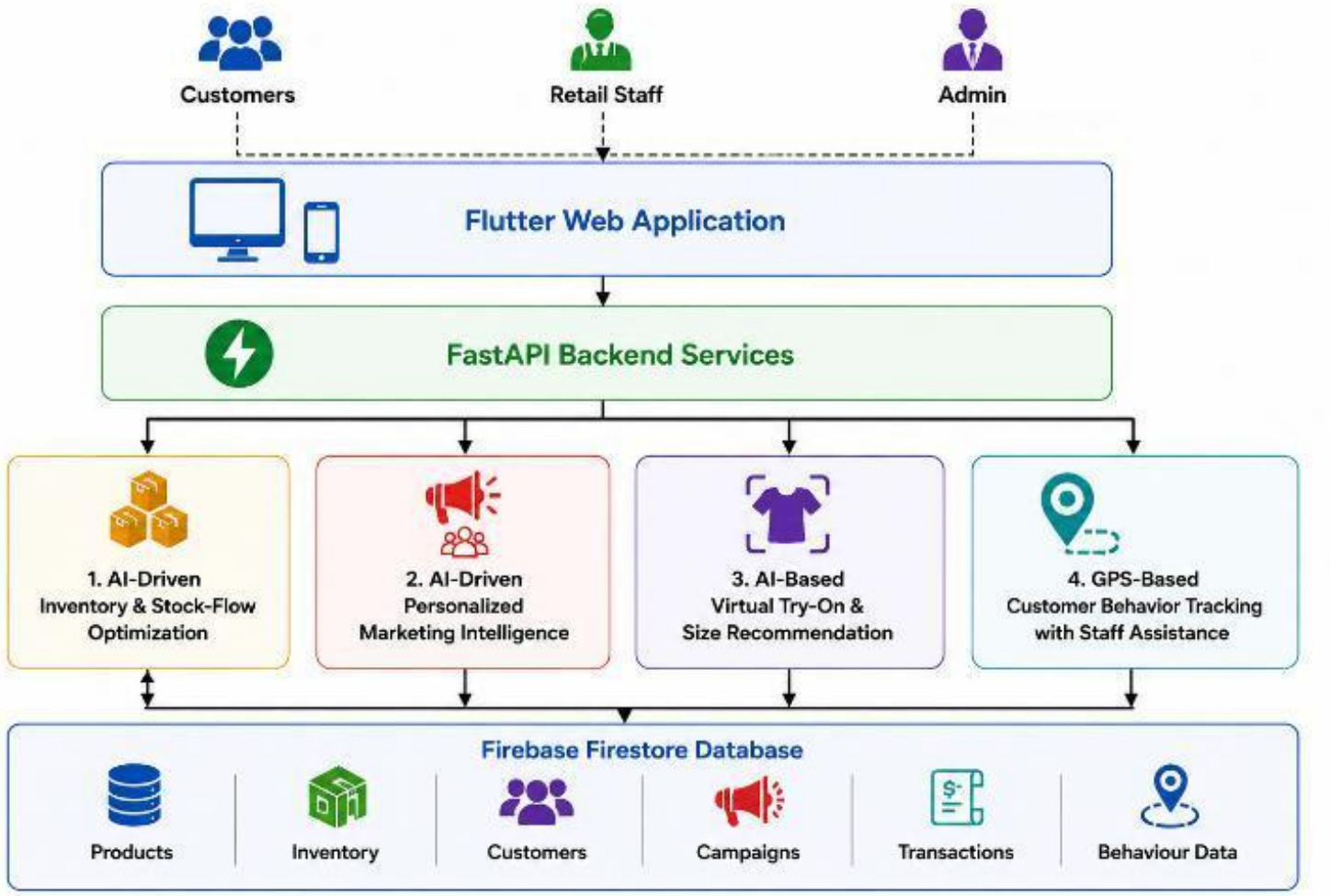
Invisible in-store behavior

Tracking normally needs beacons or cameras; staff stay reactive

Design excellence over existing solutions

Decision area	Existing solutions	Our integrated solution
Inventory	Manual or standalone forecasting; stock transfers decided by hand	XGBoost demand forecast linked to explainable, risk-scored transfer recommendations
Marketing	Broad customer segments; campaigns blind to stock levels	Redemption-probability targeting on live stock; overstock becomes a campaign
Fitting	Brand size charts; try-on separate from measurement	Measurement and brand-size prediction feed the virtual try-on in one flow
In-store service	Beacons or cameras, with staff watching manually	Phone GPS and a Random Forest intent model raise staff alerts automatically
Integration	Four separate systems with no shared data	One Firestore data model with shared IDs and a closed feedback loop
Explainability	Black-box outputs for non-technical users	Gemini plain-language explanations alongside every key prediction

Overall solution: three-layer integrated architecture



Presentation

Flutter Web — customer, staff and admin views

Application

FastAPI router per component, Firebase Auth, in-process model serving

Data

One Firebase Firestore database with shared product, customer, store and transaction IDs

External AI

Gemini and Vertex AI, called server-side only

Smart Inventory & Stock-Flow Optimization



C1 Forecasts store demand with XGBoost and recommends explainable, risk-scored stock transfers.

1	2	3	4	5	6
Load data	**Engineer features**	**Forecast demand**	**Check stock health**	**Rank source stores**	**Manager decides**
Sales history, products, promotions, stores, weather and festival calendar	Lag-1 and lag-7 demand, 7-day rolling mean, calendar and price variables	AI MODEL XGBoost predicts daily and monthly demand per product and store	Forecast vs stock gives reorder status, days-on-hand and shortages	Scores stores on stock, safety stock, distance, cost and risk	Gemini explains; approve, reject or execute; inventory updated

DATA IN sales, stock, product, store, weather, festivals

SHARES stock and overstock to C2 · in-stock products to C3

Personalized Marketing Intelligence



C2 Predicts offer redemption, promotes only in-stock products and generates the campaign poster.

1	2	3	4	5	6
Upload & clean	**Build RFM features**	**Check live stock**	**Predict redemption**	**Segment & schedule**	**Poster & explanation**
Customer, purchase and campaign CSVs cleaned and stored in Firestore	Recency, frequency and monetary value, plus product and offer attributes	Reads C1 inventory: drops out-of-stock items, flags overstock	AI MODEL Classifier scores how likely each customer is to redeem each offer	Picks high-affinity segments and times campaigns to salary cycles	Gemini Imagen poster, logo overlaid locally, plain-language reasoning

DATA IN purchase and campaign history, customers, stock from C1

SHARES redemptions and campaign data back to C1 forecasting

Virtual Try-On & Size Recommendation



C3 Predicts body measurements and brand sizes, then renders the garment on the customer's photo.

1	2	3	4	5	6
Enter profile	**Predict measurements**	**Map to brand size**	**Virtual try-on**	**Ask the AI stylist**	**Show results**
Height, weight and gender; BMI derived automatically	AI MODEL Random Forest estimates shoulder, waist and leg length	AI MODEL Random Forest converts to the right size for each brand	Photo and garment sent to Vertex AI to render the fitted look	RAG search (MiniLM + ChromaDB) over in-stock products only	Size, fitted image and styling advice; recalculates if profile changes

DATA IN height, weight, gender, photo, product catalogue, stock from C1

SHARES hosts the device for C4 and passes information to C1

GPS Behavior Tracking & Staff Assistance



C4 Classifies in-store movement from phone GPS and alerts staff, with no beacons or cameras.

1	2	3	4	5	6
Stream GPS	**Map to zones**	**Compute movement**	**Classify intent**	**Apply alert rules**	**Alert staff**
Customer app sends location every three seconds, with consent	Coordinates matched to predefined store zones	Haversine distance, velocity and direction changes	AI MODEL Random Forest: browsing or transiting; speed rule as fallback	Browsing ≥ 30 s or A→B→A pacing within 60 s; SOS skips ahead	Dashboard shows location and alert; behavior record saved

DATA IN GPS from customers' phones, store zone map, SOS requests

SHARES GPS is hosted in the application in C3

Results and testimonials

Component	Metric	Target	Achieved
C1 Inventory	Daily R^2	≥ 0.90	**0.9497**
C1 Inventory	Monthly R^2	—	**0.9353**
C2 Marketing	Accuracy (AUC)	> 55.9% baseline	**85.5% (0.934)**
C3 Try-on	Measurement MAE	≤ 2 cm	**1.12–1.45 cm**
C3 Try-on	Size accuracy	≥ 85%	**88.4%**
C4 Tracking	Intent accuracy	≥ 85%	**91.2%**

Benchmarks · Testimonials

8 / 8

integration tests passed (IT1–IT8)

User feedback / UAT: “[Add a short quote from a store manager, marketer or customer who tested the system]” — [Name, role]

Limitations and future improvements

Limitations

- Partially simulated data — main threat to external validity
- Shared collections propagate data errors
- Generative features depend on Google quotas and pricing
- Indoor GPS accuracy and battery drain
- No formal stakeholder validation yet

Future improvements

- Pilot with a partner retailer on live POS and stock data
- Hybrid forecasting with traffic and POS signals
- Larger try-on datasets and fine-tuned models
- Alternative indoor positioning alongside GPS
- Continuous learning, monitoring and a formal usability study

One connected, explainable retail brain

Stock-aware promotion · in-stock styling · behavior-to-forecast feedback — built on a modest cloud stack.

Thank you !